



THÈSE DE DOCTORAT
DE L'UNIVERSITÉ PSL

Préparée à MINES Paris

Titre de la thèse/Thesis title

Soutenue par

Pablo JEKEN RICO

Le Date de soutenance

École doctorale n°364

**Sciences Fondamentales et
Appliquées**

Spécialité

**Mathématiques Numériques,
Calcul Intensif et Données**

Composition du jury :

Prénom NOM Titre, établissement	<i>Président</i>
Prénom NOM Titre, établissement	<i>Rapporteur</i>
Prénom NOM Titre, établissement	<i>Rapporteur</i>
Prénom NOM Titre, établissement	<i>Examineur</i>
Prénom NOM Titre, établissement	<i>Examineur</i>
Prénom NOM Titre, établissement	<i>Examineur</i>
Aurélien LARCHER Mines Paris	<i>Co-encadrant de thèse</i>
Elie HACHEM Mines Paris	<i>Directeur de thèse</i>

Dedication

Acknowledgements

Contents

Contents	i
List of Figures	iii
List of Tables	v
1 Introduction	1
1.1 First Section	2
1.2 Second Section	2
2 Title of the Chapter 2	3
2.1 First Section	4
2.2 Second Section	4
2.3 Third Section	4
3 Conclusions	7
3.1 First Section	8
3.2 Second Section	8
A First Appendix	9
B Second Appendix	11

List of Figures

2.1	Mines Paris	4
2.2	Institute.	4

List of Figures

List of Tables

1.1 Table to test captions and labels. 2

Chapter 1

Introduction

Contents

1.1	First Section	2
1.2	Second Section	2

1.1 First Section

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Morbi ac iaculis neque. Sed posuere ac augue vel bibendum. Pellentesque in commodo nibh. Maecenas mattis finibus maximus. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Vestibulum vestibulum congue est ultricies interdum. Ut suscipit imperdiet ligula, eget suscipit lorem.

1.2 Second Section

This section should briefly explain the background of the study, provide a short review of the pertinent literature, state the originality or novelty of the research, and state the research objectives.

Table [1.1](#) is an example of a referenced $\text{\LaTeX}$ element.

Col1	Col2	Col2	Col3
1	6	87837	787
2	7	78	5415
3	545	778	7507
4	545	18744	7560
5	88	788	6344

Table 1.1: Table to test captions and labels.

Chapter 2

Title of the Chapter 2

Contents

2.1	First Section	4
2.2	Second Section	4
2.3	Third Section	4

2.1 First Section

Vestibulum sollicitudin, justo eget tempor volutpat, felis elit vestibulum neque, id tristique tellus dolor vitae eros. Praesent efficitur rhoncus molestie. Praesent venenatis suscipit justo, vel iaculis lectus congue ut. Suspendisse pharetra, erat ac laoreet vehicula, libero velit scelerisque libero, sit amet consequat tortor elit nec tellus. Praesent efficitur dui eu mi bibendum, nec eleifend enim iaculis. Ut et urna vehicula, volutpat magna finibus, scelerisque ante. Praesent est lorem, blandit eu nibh ut, pretium facilisis lacus. Proin gravida ligula elit, in consequat urna bibendum sed.

2.2 Second Section

Ensure that the figure will fit into either one column (Figure 2.1) or two columns (Figure 2.2). Images should be of sufficiently high resolution to be easily viewable when printed or on high resolution screens (minimum of 300 dpi).

Every figure should be cited in the text in numerical order (i.e. Figure 2.2 cannot be cited before Figure 2.1). Figures should be referred to as "Figure" not "Fig." Denote figure parts with lowercase letters (e.g. Figure 1a, Figure 1b).



Figure 2.1: Mines Paris



Figure 2.2: Institute.

2.3 Third Section

Un artículo científico se sustenta en trabajo previo y por lo tanto el uso de referencias es muy importante. Las citas en nuestra revista se harán de acuerdo al uso de bibtex

integrado en LaTeX. En el archivo referencias se presentan algunos ejemplos típicos que se usa en nuestra literatura. Para citar alguna referencia que se encuentra en nuestro archivos referencia solamente hay que usar `\cite{Referencia}`, por ejemplo una cita al profesor R. Sagar [?] o la cita a un libro [?]. Con bibtex no tenemos que preocuparnos por la numeración de las referencias. Por lo tanto solamente tenemos que preocuparnos por escribir las referencias en el formato mostrado en el archivo `Biblio.bib`.

Chapter 3

Conclusions

Contents

3.1	First Section	8
3.2	Second Section	8

3.1 First Section

In research articles, the materials and methods used in the study should be described together—first the materials, and then the methods. Enough information should be provided to enable repetition of the research. For commercial sources of the materials, the name of the company, and the town and country in which they are headquartered should be indicated. To avoid an excessively long methods section, methods that have already been published should be indicated with a reference, with only the relevant modifications described.

3.2 Second Section

Add your theoretical background here. Some example text: As we learned from our textbook, Fourier introduced the concept of representing a given function $f(x)$ by a trigonometric series of sines and cosines:

$$f(x) = \frac{a_0}{2} + \sum_{i=1}^{\infty} (a_n \cos nx + b_n \sin nx) \quad x \in (-\pi, \pi]. \quad (3.1)$$

You can reference numbered equations, figures, tables, algorithms, and code like this: Equation [3.1](#), etc.

Appendix A

First Appendix

Appendix B

Second Appendix

RÉSUMÉ

Sed tamen haec cum ita tutius observentur, quidam vigore artuum inminuto rogati ad nuptias ubi aurum dextris manibus cavatis offertur, inpigre vel usque Spoletium pergunt. haec nobilium sunt instituta.

MOTS CLÉS

Caesar licentia post honoratis haec adhibens urbium honoratis nullum Caesar.

ABSTRACT

Ego vero sic intellego, Patres conscripti, nos hoc tempore in provinciis decernendis perpetuae pacis habere oportere rationem. Nam quis hoc non sentit omnia alia esse nobis vacua ab omni periculo atque etiam suspicione belli?

KEYWORDS

Delatus delatus nominatus onere aut trahebatur quod tenus et bonorum.